

SonarQube Quality Gate & Code Audit Report

Project Name: **Notes App**
Assessment Type: Automated Testing & Continuous Integration Audit

1. Project Architecture & Overview

The target application, '**Notes App**', is a full-stack JavaScript web application comprising two primary modules:

- **Project Goal:** Full-stack Notes application for creating, editing, and managing user notes.
- **Core Stack:** **React 18** frontend and **Node.js / Express 5** backend with **MongoDB**.
- **Test Setup:** Tested via **Jest** (frontend) and **Mocha / Chai / NYC** (backend) with combined SonarQube reporting.

2. Executive Summary

The latest SonarQube audit for the Notes Application confirmed a **PASSED** Quality Gate status. By implementing unified unit testing and coverage pipelines across both the frontend (`react-scripts test --coverage`) and backend (`nyc --reporter=lcov mocha`), code coverage successfully rose from 0.0% to 91.5% across 556 lines to cover. The codebase achieved perfect 'Grade A' ratings in Security, Reliability, and Maintainability with zero open bugs or code smells.

3. Detailed SonarQube Metrics:

Metric Category	Rating	Value / Count	Audit Details
Quality Gate	PASSED	Green Status	Meets all organizational release requirements
Security	A	0 Open Issues	Zero vulnerabilities identified in frontend or backend code
Reliability	A	0 Open Issues	Zero software bugs or unhandled promise rejections
Maintainability	A	6 Open Issues	Minor Review Needed
Accepted Issues	—	0 Issues	No security or quality risks deferred
Code Coverage	OPTIMAL	91.5% 556 lines cover	Automated test execution linked via lcov.info reports
Duplications	A	0.0% (0 / 12k lines)	Zero redundant or duplicate code blocks across ~10k lines
Security Hotspots	A	0 Hotspots	No sensitive code areas flagged for manual security review

4. Screen Short:

localhost:9000/dashboard?id=Notes-app-report&codeScope=overall

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

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Analysis

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Issues

Security h... **Deprecated**

Reporting

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Activity

Quality gate hist... **New**

Policies

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Notes app report / Overview

Overview

3.6k Lines of Code · Version not provided · Last analysis 3 minutes ago

[Set as homepage](#)

Quality gate

Passed

The last analysis has warnings. [See details](#)

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New Code Overall Code

Security	Reliability	Maintainability
0 Open issues	0 Open issues	6 Open issues
Accepted issues	Coverage	Duplications
0	91.5%	0.0%
Valid issues that were not fixed	On 564 lines to cover.	On 12k lines.
Security Hotspots Deprecated		
0		

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Overview

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Graph type

Issues

There isn't enough data to generate an activity graph.

Aug 29, 2026, 12:56 AM

Quality gate: Passed

not provided

First analysis: 6 Issues • 91.5% Coverage • 0.0% Duplications

[See full history of analyses](#)

Unit testing report:

File	% Stmts	% Branch	% Funcs	% Lines
All files	93.57	97.56	100	93.57
backend	100	100	100	100
app.js	100	100	100	100
backend/config	100	100	100	100
logger.js	100	100	100	100
backend/controllers	89.23	100	100	89.23
authController.js	93.1	100	100	93.1
notesController.js	86.11	100	100	86.11
backend/middleware	92.59	92.85	100	92.59
authMiddleware.js	88.23	87.5	100	88.23
errorHandler.js	100	100	100	100
loggerMiddleware.js	100	100	100	100
backend/models	100	100	100	100
Note.js	100	100	100	100
User.js	100	100	100	100
backend/routes	100	100	100	100
authRoutes.js	100	100	100	100
notesRoutes.js	100	100	100	100
backend/utils	100	100	100	100
generateToken.js	100	100	100	100